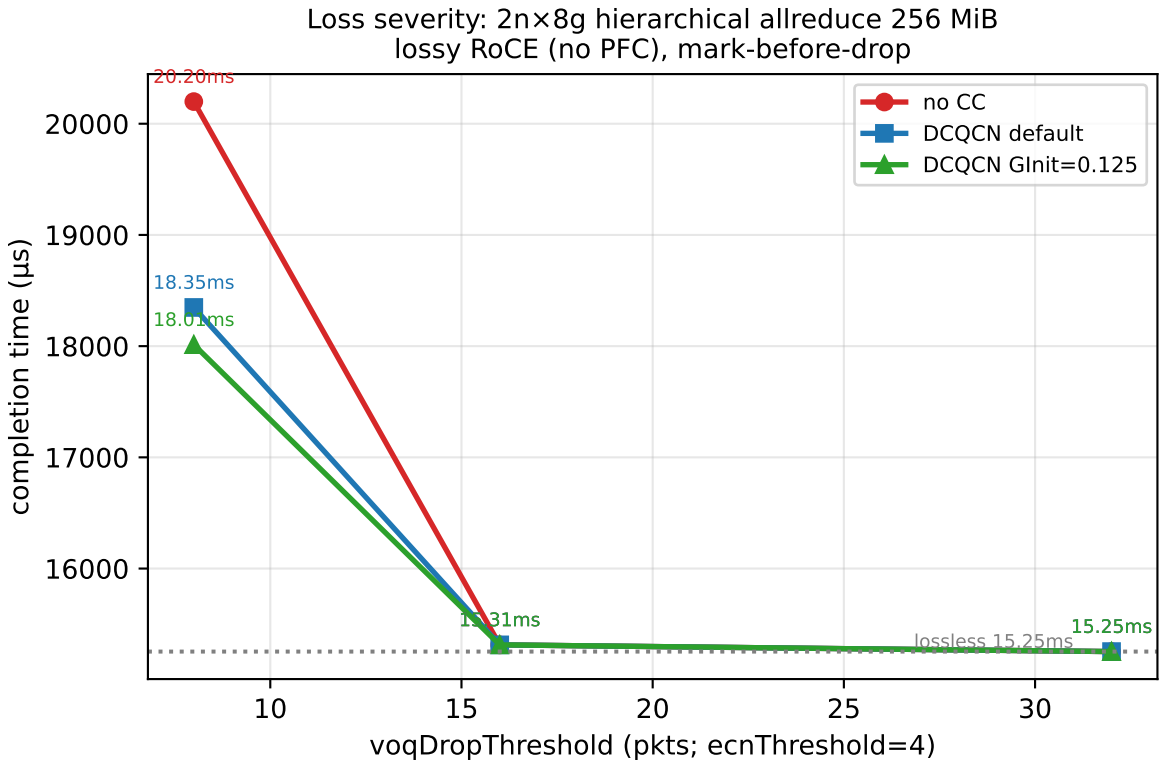
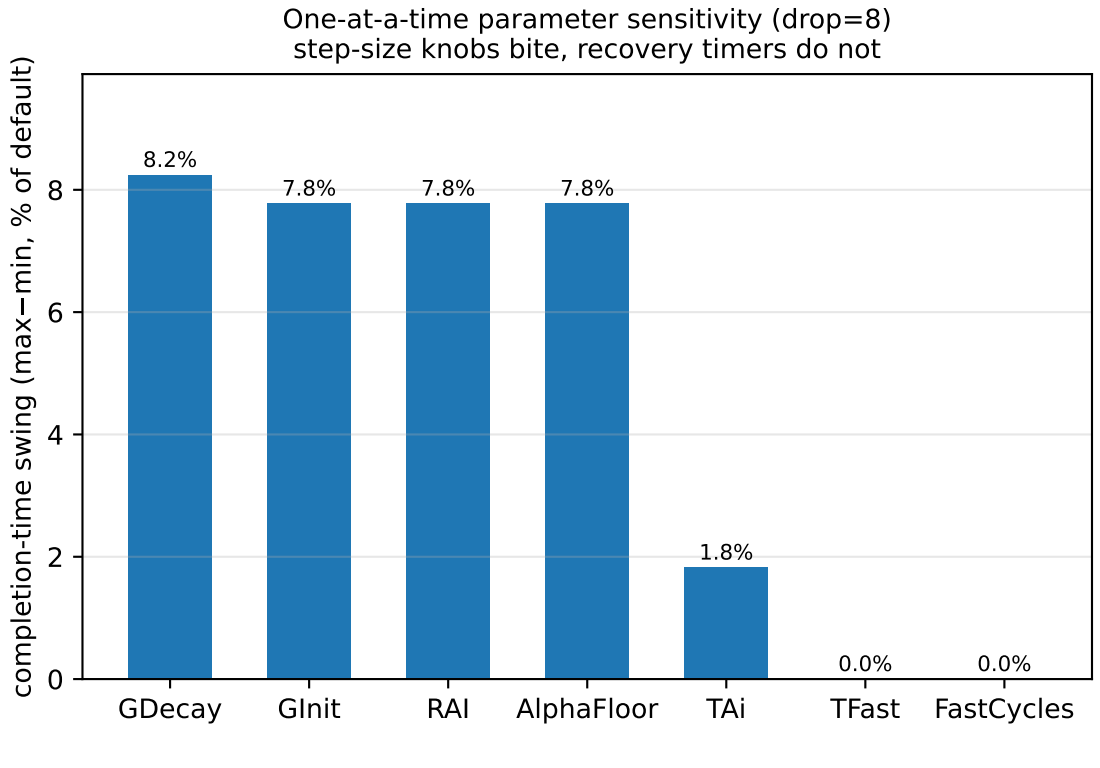
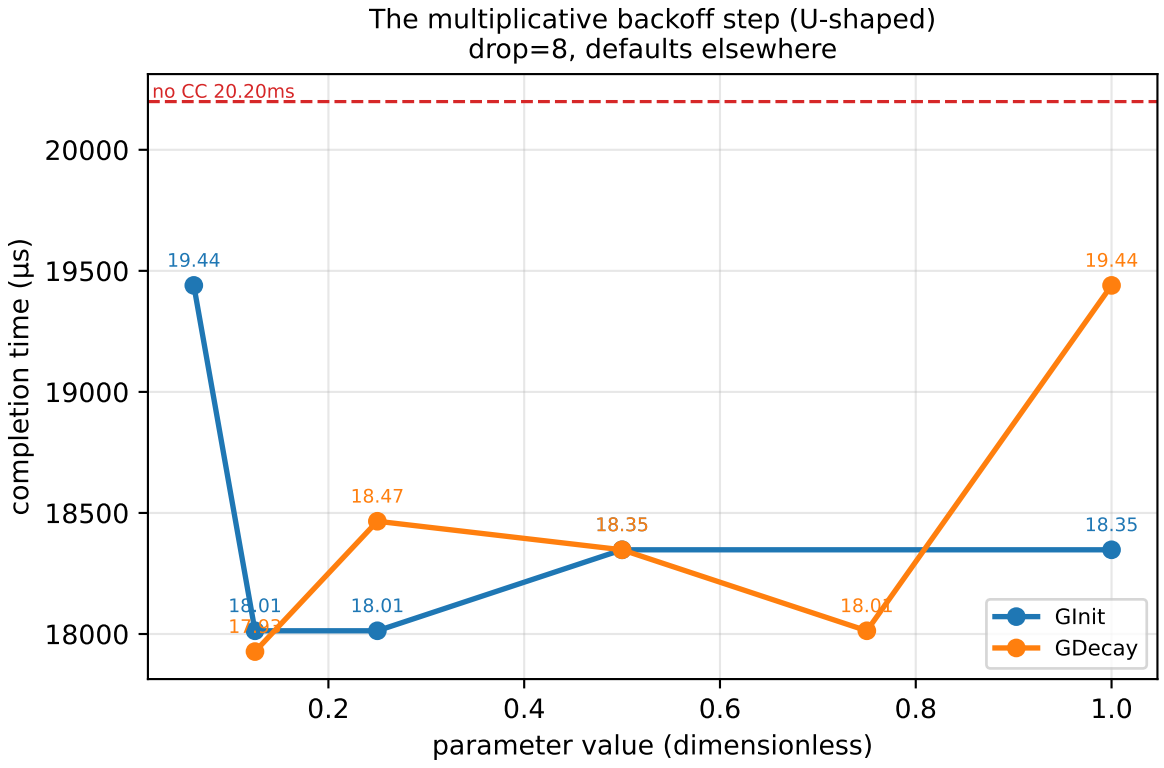
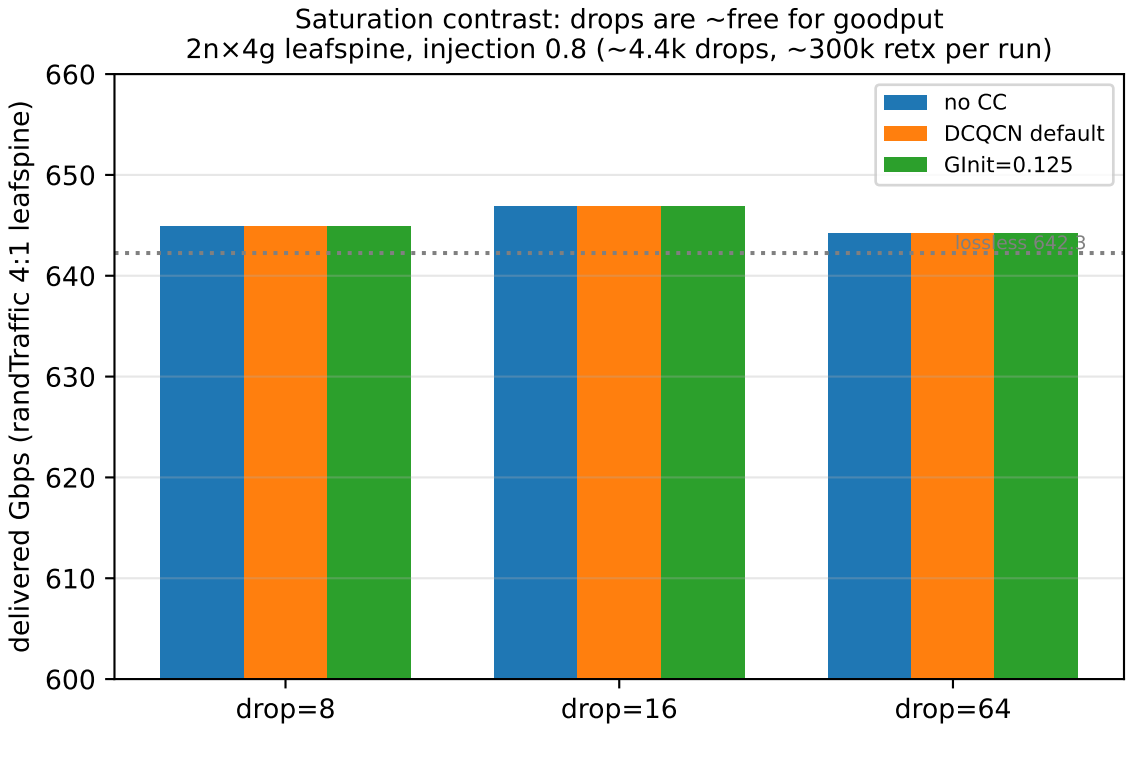

Loss severity: 2n×8g hierarchical allreduce 256 MiB
lossy RoCE (no PFC), mark-before-drop
no CC
DCQCN default
DCQCN GInit=0.125
20.20ms
18.35ms
18.01ms
15.31ms
15.25ms
lossless 15.25ms
completion time (µs)
voqDropThreshold (pkts; ecnThreshold=4)


One-at-a-time parameter sensitivity (drop=8)
step-size knobs bite, recovery timers do not
completion-time swing (max−min, % of default)
8.2%
7.8%
7.8%
7.8%
1.8%
0.0%
0.0%
GDecay
GInit
RAI
AlphaFloor
TAi
TFast
FastCycles


The multiplicative backoff step (U-shaped)
drop=8, defaults elsewhere
no CC 20.20ms
19.44
18.47
18.35
18.35
18.01
18.01
18.01
19.44
GInit
GDecay
completion time (µs)
parameter value (dimensionless)


Saturation contrast: drops are ~free for goodput
2n×4g leafspine, injection 0.8 (~4.4k drops, ~300k retx per run)
no CC
DCQCN default
GInit=0.125
lossless 642.3
delivered Gbps (randTraffic 4:1 leafspine)
drop=8
drop=16
drop=64